UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational A • 2023

GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

B01. In which eukaryotic organelle would the enzymes for degradation of damaged organelles or breakdown of food substance from food vacuoles be found?
A) mitochondria
B) rough endoplasmic reticulum
C) peroxisome
D) nucleus
E) lysosome

B02. In Mendelian classical genetics, the genotype Pp would be referred to as
A) homozygous.
B) homozygous dominant.
C) homozygous recessive.
D) heterozygous.
E) codominant.

B03. The process of reading a messenger RNA sequence and converting it into a protein sequence is called
A) replication.
B) regulation.
C) transduction.
D) transcription.
E) translation.

B04. Which of the following organic macromolecules is best characterized as nonpolar, hydrophobic, and found in fats and oils.
A) phospholipids
B) triglycerides
C) amino acids
D) nucleotides
E) monosaccharides

B05. The type of cell division that produces gametes is called
A) sexual reproduction.
B) mitosis.
C) meiosis.
D) fusion.
E) budding.

B06. Male birds performing intricate dancing sequences to win the interest of same-species female birds is known as
A) genetic drift.
B) sexual selection.
C) gene flow.
D) speciation.
E) hybridization.

B07. The major photosynthetic pigment that causes leaves on deciduous trees to be green is
A) chlorophyll.
B) carotenoid.
C) anthocyanin.
D) phycobillin.
E) flucoxanthin.

B08. Epithelial cells in the skin that produce pigment are called
A) keratinocytes.
B) basal cells.
C) squamous cells.
D) melanocytes.
E) mast cells.

B09. The organ system responsible for excretion of nitrogenous wastes from the human body and fluid balance is the
A) Musculoskeletal system.
B) Urinary system.
C) Reproductive system.
D) Digestive system.
E) Nervous system.

B10. Organisms in the same ____ would always be found in the same ______.
A) Domain; Kingdom
B) Kingdom; species
C) Genus; species
D) Family; Order
E) Class; Order

B11. All of the following are examples of unity in diversity except
A) all organisms are made of cells.
B) ribosomes are the cellular machinery that makes proteins for all organisms.
C) the wings of butterflies and the wings of birds are homologous structures.
D) the forelimbs of mammals (cats, bats, whales, humans) have similar structures.
E) inherited information is stored in DNA molecules for all living organisms.

B12. The process of ______ is directly linked to, and occurs before, cell division.
A) DNA replication
B) translation
C) gene expression
D) transcription
E) respiration

B13. The alleles in the following genetic cross are incompletely dominant. What percentage of the progeny will express the intermediate phenotype?

Aa x aa

A) 0%
B) 25%
C) 50%
D) 75%
E) 100%

B14. Which of the following is correctly paired with its binding site?
A) RNA polymerase - activator sequence
B) Ribosome - promoter
C) Initiator tRNA - stop codon
D) Repressor - operator
E) DNA polymerase - +1

B15. Gram-positive bacteria are differentiated from Gram-negative bacteria by
A) the presence of a thicker layer of peptidoglycan in Gram-positive bacteria.
B) Gram-positive bacteria staining purple in the Gram stain process.
C) the presence of only one membrane, the cytoplasmic membrane, in Gram-positive bacteria.
D) the absence of lipopolysaccharide (LPS) in Gram-positive bacteria.
E) all of the above.

B16. In biological hierarchy, which level immediately precedes tissues moving from smallest to largest?
A) cells
B) organ systems
C) organelles
D) organs
E) atoms

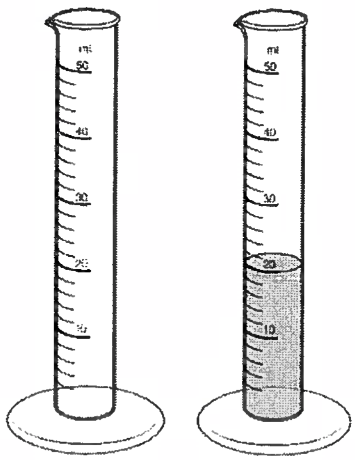
B17. In October 2022, the Centers for Disease Control and Prevention issued a Food Safety Alert and the store recalled frozen falafel due to contamination with
A) Hepatitis virus.
B) *Salmonella.*
C) *Listeria.*
D) Coronavirus.
E) Shiga-toxin producing *Escherichia coli.*

B18. Cellular components that catalyze chemical reactions and are made of amino acids are specifically called
A) inorganic catalysts.
B) organelles.
C) enzymes.
D) ribozymes.
E) proteins.

B19. In a population at Hardy-Weinberg equilibrium, 302 organisms out of 491 express the dominant phenotype. What percent of the population are homozygous recessive?
A) 29.1%
B) 38.5%
C) 61.5%
D) 62.0%
E) 78.4%

B20. The reaction in the nitrogen cycle that reduces nitrates completely to dinitrogen gas through a nitrite intermediary is called
A) denitrification.
B) nitrification.
C) ammonification.
D) nitrogen fixation.
E) anammox.

C01. If the empty 50 mL graduated cylinder shown below weighs 177.0 grams and the one with the liquid weighs 204.0 grams, what is the density of the liquid in the graduated cylinder?



A) 1.35 g/mL
B) 1.00 g/mL
C) 1.12 g/mL
D) 10.2 g/mL
E) 0.74 g/mL

C02. When the following equation is balanced using the smallest whole number coefficients, what is the sum of the coefficients?

$Al(NO_3)_3(aq) + Na_3PO_4(aq) \rightarrow AlPO_4(s) + NaNO_3(aq)$

A) 3
B) 4
C) 5
D) 6
E) 8

C03. How many electrons are in a neutral atom of Carbon-14?

A) 4
B) 6
C) 8
D) 12
E) 14

C04. What type of bond holds the oxygen and hydrogen atoms together in a water molecule?

A) Ionic bond
B) Covalent bond
C) Hydrogen bond
D) Covalent and oxygen bonds
E) Covalent and hydrogen bonds

C05. If 0.100 moles of neon gas are placed in a 2.00 liter rigid container at 25.0 °C and then the container is heated to 190.0 °C, what will the final pressure of the gas inside the container be?

A) 1.22 atm
B) 0.78 atm
C) 1.56 atm
D) 3.80 atm
E) 1.90 atm

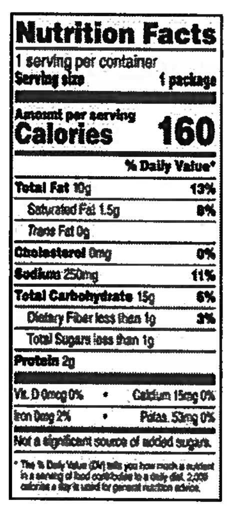
C06. What type of intermolecular forces would you expect to find in liquid methane, CH_4?

A) Ion-ion forces
B) Hydrogen bonding
C) Dipole-dipole forces
D) Dispersion Forces
E) None

C07. The amount of heat released or absorbed by a chemical reaction that occurs at constant pressure is called the

A) internal energy
B) enthalpy of reaction
C) entropy of reaction
D) spontaneous energy
E) reaction deficit energy

C08. A 1 oz snack bag of Cheetos contains 160 Calories of energy. How many grams of ice could be melted using the energy stored in one serving of Cheetos?



A) 2000 g
B) 2.00 g
C) 534 g
D) 479 g
E) 669 g

C09. What is the equilibrium constant K_P for the gas-phase reaction

$$2A + B \rightleftharpoons 3C + D$$

if the gas pressures at equilibrium are found to be

Gas	Equilibrium Pressure (atm)
A	0.50
B	0.25
C	0.10
D	0.15

A) 2.4×10^{-3}
B) 420
C) 0.12
D) 8.3
E) 1.2×10^{-3}

C10. What is the pH of 300.0 mL of 0.015 M hydrochloric acid?

A) 0.300
B) 0.500
C) 1.35
D) 1.82
E) 2.00

C11. Which of these is the correct equation for calculating the K_{sp} of $Ca(OH)_2$ from the molar solubility, x?

A) $K_{sp} = 16x$
B) $K_{sp} = x^2$
C) $K_{sp} = 4x^3$
D) $K_{sp} = 27x^4$
E) $K_{sp} = 108x^5$

C12. Which of these is a possible product of a chemical reaction in which N_2 gas undergoes reduction?

A) N atoms
B) NO_2
C) NH_3
D) N_2O_4
E) N_2O

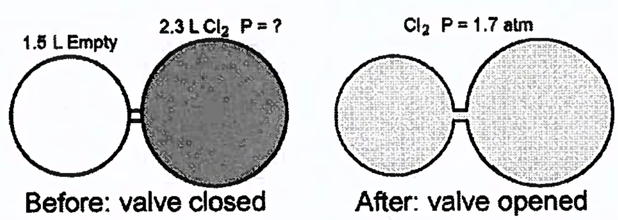
C13. The rate law for the chemical reaction between A and B is first order with respect to A and also first order with respect to B. Which of the following statements is true?

A) Sometimes A reacts first in the reaction and sometimes B reacts first.
B) Doubling the concentration of A will double the rate of the reaction.
C) Doubling the concentrations of both A and B will double the rate of the reaction.
D) Changing the concentrations of A and B will not change the rate of the reaction.
E) The rate law cannot be first order with respect to both A and B. One must be first and the other second.

C14. How many atoms are in a ball of copper wire that has a mass of 3.45 grams?

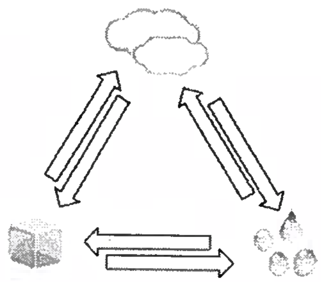
A) 3.27×10^{22}
B) 8.74×10^{22}
C) 3.27×10^{24}
D) 8.74×10^{24}
E) 8.27×10^{23}

C15. A 1.5 L bulb and a 2.3 L bulb are connected by a valve. The larger bulb contains Cl_2 gas at an unknown pressure, and the smaller bulb is empty. When the valve is opened, the pressure of the Cl_2 gas in both bulbs is 1.7 atm. What was the original Cl_2 pressure in the 2.3 L bulb? Assume the volume of the valve tube is negligible.



A) 2.8 atm
B) 0.36 atm
C) 1.7 atm
D) 1.0 atm
E) 3.3 atm

C16. Given this diagram and the information on the data sheet, what is the heat of sublimation for water?



A) 6.766 J/g
B) 1926 J/g
C) 2260 J/g
D) 2928 J/g
E) 2594 J/g

C17. How many of the following names and formulas do not match correctly?

Chemical Formula	Chemical Name
$Cr_2(SO_4)_3$	chromium(III) sulfate
CS_2	carbon disulfide
$(NH_4)_2CO_3$	ammonia carbon trioxide
$FeCl_2$	iron chloride
NI_3	nitrogen iodide

A) 1
B) 2
C) 3
D) 4
E) 5

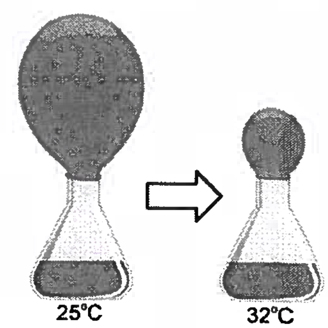
C18. Which of these is a valid set of quantum numbers?

A) $n = 1$ $\ell = 1$ $m_\ell = 0$ $m_s = -½$
B) $n = 2$ $\ell = 1$ $m_\ell = -1$ $m_s = +½$
C) $n = 3$ $\ell = 2$ $m_\ell = +2$ $m_s = +⅓$
D) $n = 4$ $\ell = 3$ $m_\ell = +3$ $m_s = -⅓$
E) $n = 5$ $\ell = 4$ $m_\ell = +5$ $m_s = +½$

C19. A 50.0 mL sample of 0.100 M HNO_3 is titrated to the equivalence point using 10.0 mL of 0.500 M NaOH. What is the final pH of the solution?

A) 1
B) 5
C) 7
D) 10
E) 60

C20. What is the sign on the change in internal energy (ΔU) for the chemical reaction carried out in a balloon-covered flask shown below?



A) $\Delta U > 0$
B) $\Delta U < 0$
C) $\Delta U = 0$
D) $\Delta U < 0$ at the top and $\Delta U > 0$ at the bottom
E) More information is needed to answer the question

Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	9	10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp} = 0°C
T_{bp} = 100°C
c_{ice} = 2.09 J/g·K
c_{water} = 4.184 J/g·K
c_{steam} = 2.03 J/g·K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/*m*
K_b = 0.512 °C/*m*

Energy Unit Conversions

1 Calorie = 1000 calories
1 calorie = 4.184 J

Constants

R = 0.08206 L·atm/mol·K
R = 8.314 J/mol·K
R = 62.36 L·torr/mol·K
e = 1.602×10^{-19} C
N_A = 6.022×10^{23} mol^{-1}
k = 1.38×10^{-23} J/K
h = 6.626×10^{-34} J·s
c = 3.00×10^{8} m/s
R_H = 2.178×10^{-18} J
m_e = 9.11×10^{-31} kg

Other Data

All other necessary data for this exam is included in the problems themselves.

P01. According to Feynman, quantum electrodynamics (QED) describes all physical phenomena except two. Which two phenomena does QED not describe?
A) gravity and radioactivity
B) gravity and light
C) radioactivity and light
D) light and chemistry
E) chemistry and radioactivity

P02. According to Feynman, a __________ is an instrument that can detect a single photon.
A) photodiode
B) photomultiplier
C) phototransistor
D) cathode ray tube
E) Geiger counter

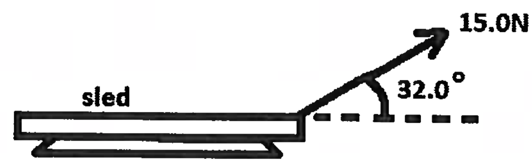
P03. According to Feynman, iridescence is the production of colors by the partial reflection of white light by two surfaces. An example of iridescence is…
A) the deep colors of stained glass
B) the rich colors of lead-based paints
C) the bright colors of cloth dyes
D) the saturated colors of television pixels
E) the brilliant colors of peacocks

P04. Though not advisable, if you did look at the Sun, what region of the Sun would you see as the disk of the Sun?
A) the Corona
B) the Chromosphere
C) the Photosphere
D) the Convection Zone
E) the Solar Wind

P05. A cube of cheese that is exactly 27.0mm on a side is melted into a circular puddle that has a diameter of 22.0cm. What is the thickness of the circular puddle of cheese?
A) 0.13 mm
B) 0.41 mm
C) 0.52 mm
D) 1.6 mm
E) 2.1 mm

P06. A small truck was initially travelling at 20.0m/s. The truck accelerated at a constant rate up to a speed of 35.0m/s. If the acceleration took a total of 6.00seconds, then how far did the truck travel while it was accelerating?
A) 45.0 m
B) 90.0 m
C) 120 m
D) 165 m
E) 210 m

P07. A sled with a mass of 7.00 kg is pulled across frictionless snow by a force of 15.0 N. The force is directed upwards at an angle of 32.0° above the horizontal (as shown). What is the horizontal acceleration of the sled?



A) 1.14 m/s^2
B) 1.82 m/s^2
C) 2.14 m/s^2
D) 2.96 m/s^2
E) 3.96 m/s^2

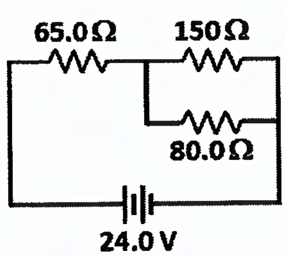
P08. A child starts from rest at the top of a large slide, at which point her center of mass is 4.30m above the ground. Ignoring friction and air resistance, how fast is the child moving when she reaches the bottom of the slide? Note: At the bottom of the slide, the child's center of mass is 0.80m above the ground.
A) 3.96 m/s
B) 5.86 m/s
C) 6.49 m/s
D) 8.28 m/s
E) 9.18 m/s

P09. A meter stick is arranged horizontally with a fulcrum (pivot point) located exactly at the 50.0cm mark (the center of mass of the meter stick). A 60.0g mass is attached to the meter stick at the 30.0cm mark. At what location on the meter stick should a 100.0g mass be attached to balance the meter stick?
A) the 12.0cm mark
B) the 18.0cm mark
C) the 62.0cm mark
D) the 68.0cm mark
E) the 82.0cm mark

P10. A 250.0g mass is attached to a vertically hanging spring that has a spring constant of 140N/m. The mass is pulled 15.0cm from the equilibrium position and released from rest, resulting in harmonic motion. What is the speed of the mass as it passes through the equilibrium point during each oscillation?
A) 3.55 m/s
B) 15.8 m/s
C) 23.7 m/s
D) 39.4 m/s
E) 84.0 m/s

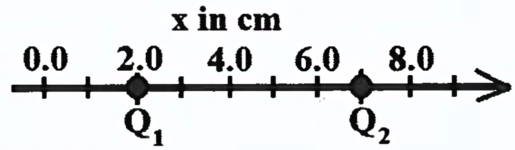
P11. A perfect Carnot engine absorbs heat energy from a 560K hot reservoir, and exhausts heat energy into a 220K cold reservoir. What is the efficiency of the Carnot engine?
A) 16%
B) 25%
C) 39%
D) 49%
E) 61%

P12. Given the following circuit, determine the current flowing in the 65.0Ω resistor.



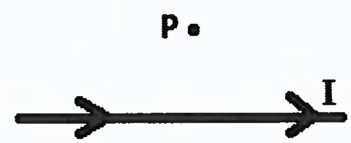
A) 81.4 mA
B) 205 mA
C) 369 mA
D) 590 mA
E) 829 mA

P13. Two charges, $Q_1 = 85.0nC$ located at x = 2.0cm, and $Q_2 = -63.0nC$ located at x = 7.0cm, are positioned on the x-axis as shown. What is the magnitude of the electric potential at the origin (x = 0.0cm) due to the presence of these two charges?



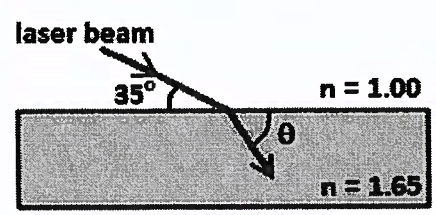
A) 18.0 kV
B) 20.3 kV
C) 30.1 kV
D) 38.5 kV
E) 46.4 kV

P14. A long straight wire carries a current of 2.25A, directed from left to right (as shown). What is the magnetic field at the point P, located 4.00cm above the wire?



A) 11.3 μT out of the page
B) 11.3 μT into the page
C) 28.1 μT out of the page
D) 28.1 μT into the page
E) 35.5 μT out of the page

P15. A laser beam travelling in air is directed onto a rectangular prism of glass, and is refracted as shown. The index of refraction of the glass is 1.65. Determine the indicated angle, θ, of the beam in the glass.



A) 20.3°
B) 29.8°
C) 45.4°
D) 60.2°
E) 71.2°

P16. A convex mirror with a radius of curvature of 28.0cm is located 11.0cm to the right of a light bulb. What is the magnification of the image of the light bulb?

A) 4.67
B) 3.16
C) 1.65
D) 0.72
E) 0.56

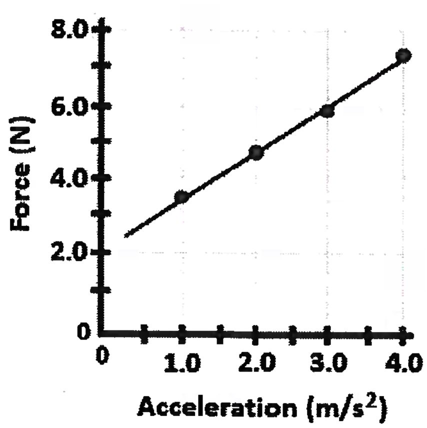
P17. A hydrogen atom undergoes a transition from the $n = 9$ state to the $n = 4$ state. What is the wavelength of the photon emitted during this transition?

A) 656 nm
B) 820 nm
C) 1220 nm
D) 1460 nm
E) 1820 nm

P18. A subatomic particle that is classified as a baryon, such as a proton, is composed of ____

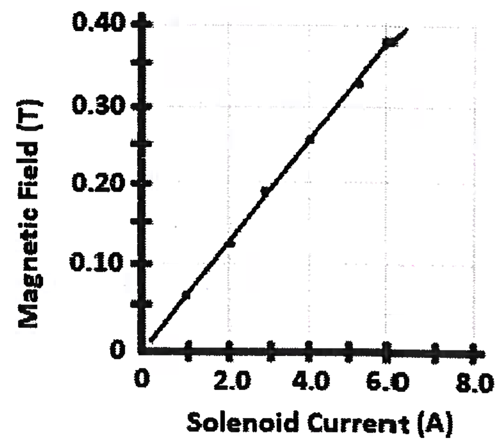
A) three quarks
B) two quarks
C) a quark and an anti-quark
D) a quark and two anti-quarks
E) two quarks and an anti-quark

P19. Shown below is a graph of the force acting on a toy car as a function of the acceleration of the car. Based on these data, what is the approximate mass of the toy car?



A) 0.75 kg
B) 1.3 kg
C) 1.8 kg
D) 2.0 kg
E) 2.7 kg

P20. The magnetic field inside a solenoid is measured as the current through the solenoid is varied. The data are plotted below. Given that the solenoid is linear and 2.00cm long, determine the approximate number of turns of wire that make up the solenoid.



A) 300 turns
B) 1000 turns
C) 3000 turns
D) 10,000 turns
E) 50,000 turns

UIL Science Contest Answer Sheet

Conference ______ Grade Level ______ Contestant # ________

Biology		Chemistry		Physics	
B01	______	C01	______	P01	______
B02	______	C02	______	P02	______
B03	______	C03	______	P03	______
B04	______	C04	______	P04	______
B05	______	C05	______	P05	______
B06	______	C06	______	P06	______
B07	______	C07	______	P07	______
B08	______	C08	______	P08	______
B09	______	C09	______	P09	______
B10	______	C10	______	P10	______
B11	______	C11	______	P11	______
B12	______	C12	______	P12	______
B13	______	C13	______	P13	______
B14	______	C14	______	P14	______
B15	______	C15	______	P15	______
B16	______	C16	______	P16	______
B17	______	C17	______	P17	______
B18	______	C18	______	P18	______
B19	______	C19	______	P19	______
B20	______	C20	______	P20	______

B Score ☐ **C Score** ☐ **P Score** ☐

Grader Initials ______ ______ ______ **OVERALL SCORE** ☐